

COM410 Programming in Practice

A3. Abstract Data Types

CHALLENGE

Classroom

Create a new Java project called **Classroom** and populate with a class called **ArrayBag** that provides storage and functionality for a collection of String objects. In the **main()** method of the **Classroom** class, provide the following functionality.

1. Create an array of 10 student first names.
2. Populate the bag with 100 random selections from the list of names and display the bag contents on the console.
3. Find the name most frequently held in the bag and report it and its number of occurrences. If there are multiple names with the joint-top number of occurrences, report all of the jointly most-popular names.
4. Remove all occurrences of these names from the bag and display the number of names that remain in the bag.
5. Generate a table of all remaining names with the number of times that they appear. Only names that appear in the bag at least once should be included.

Please upload a screenshot of your solution running to the A3 Challenge submission link on Blackboard, by the deadline shown there.